

The Sampling Theorem With Constant Amplitude Variable Width Pulses

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Abstract—This paper proves a novel sampling theorem with constant amplitude and variable width pulses. The theorem states that any bandlimited baseband signal within ± 0.637 can be represented by a pulsewidth modulation (PWM) waveform with unit amplitude. The number of pulses in the waveform is equal to the number of Nyquist samples and the peak constraint is independent of whether the waveform is two-level or three-level. The proof of the sampling theorem uses a simple iterative technique that is guaranteed to converge to the exact PWM representation whenever it exists. The paper goes on to develop a practical matrix based iterative technique to generate the PWM waveform that is guaranteed to converge exponentially. The peak constraint in the theorem is only a sufficient condition. In fact, many signals with higher peaks, e.g., lower than Nyquist frequency sinusoids, can be accurately represented by a PWM waveform.

Index Terms—Bandlimited signal, intersymbol interference, pulsewidth modulation, sampling theorem, switching amplifier.

I. INTRODUCTION

THE classic Nyquist sampling theorem [1] states that a bandlimited baseband signal $x(t)$ within bandwidth B can be exactly recovered from its sample values by low-pass filtering if the samples are taken at a rate higher than $2B$. This paper proves that pulsewidth modulation (PWM) waveforms can also represent bandlimited signals with zero distortion. Instead of representing a baseband signal by a series of sample heights, this theorem represents a signal as a series of pulses with constant amplitude and variable widths. Low-pass filtering can recover the original signal exactly. Fig. 1 illustrates the basic idea. In the upper branch, the digital samples are represented as a series of impulses and in the lower branch, the samples are converted into pulsewidths and then represented as a PWM waveform. Low-pass filtering both the conventional pulses in the upper branch and the PWM waveform in the lower branch results in identical signals. Achieving linearity is easier using the PWM system because of the good time resolution obtainable in modern circuitry [2], [3]. The characteristic of constant amplitude in the PWM representation also allows the use of switching amplifiers, which are very power efficient. In fact, the *raison d'être* for these results is power digital to analog conversion [4].

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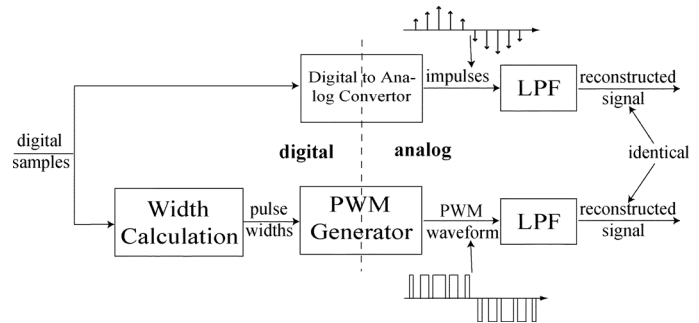


Fig. 1. System block diagram.

PWM has, however, become almost universal and the technique presented in this paper has the potential to enhance its performance wherever it is used.

The basic result of this paper shows that any bandlimited baseband signal $x(t)$ within bandwidth B and $|x(t)| \leq 2/\pi = 0.637$, can be represented by a PWM waveform with frequency $2B$ and voltage ± 1 . By passing the PWM waveform through an ideal low-pass filter with cutoff frequency B , we will get back the exact original bandlimited signal. Moreover, this paper shows that the peak constraint 0.637 is tight in the sense that for any value greater than 0.637, there exist signals which cannot be represented by a PWM waveform exactly. However, this peak constraint is just a sufficient condition, and definitely not necessary. Many signals with higher peaks can also be exactly represented by PWM waveforms.

This paper proves the sampling theorem using a particular class of PWM waveforms whose pulses are centered at the sampling times, and examines both two-level and three-level PWM waveforms. Fig. 2 illustrates the PWM representation of a bandlimited baseband signal for both the two-level and the three-level cases. For this class of PWM waveforms, this paper provides practical techniques to calculate the pulsewidths for the distortion free PWM waveform whenever it exists, and the peak constraint of 0.637 is independent of whether the waveform is two-level or three-level.

Both two-level and three-level PWM are widely used in many applications. Fig. 3 illustrates two-level PWM, e.g., a single ended load fed from a push-pull stage. Fig. 4 shows an example of three-level PWM implemented by feeding a differential load with two separate push-pull stages. The three-level PWM waveform is generated as the difference between two two-level waveforms, and Table I is its truth table. In these circuits, the transistors are either saturated or turned off nearly all of the time. Hence, the power dissipation is low. These figures assume that the loads incorporate an intrinsic low-pass filter. Many loads

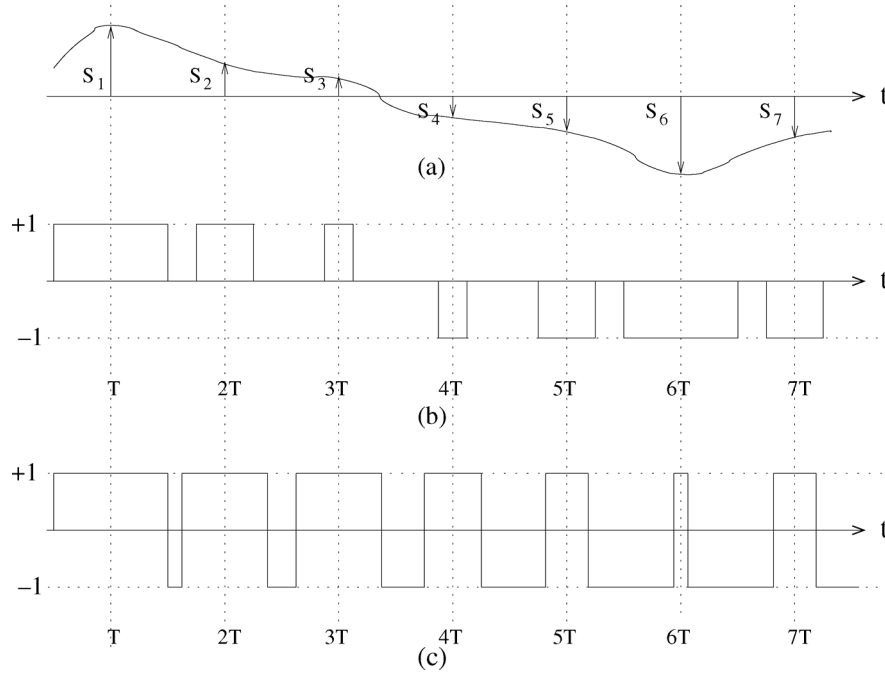


Fig. 2. PWM representation. (a) Baseband signal. (b) Three level. (c) Two level.

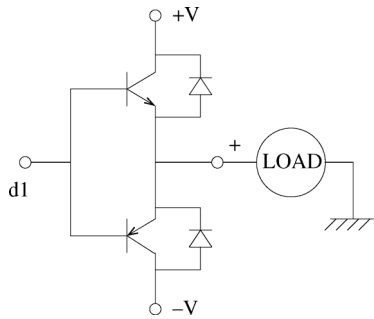


Fig. 3. Two-level PWM implementation (half-bridge).

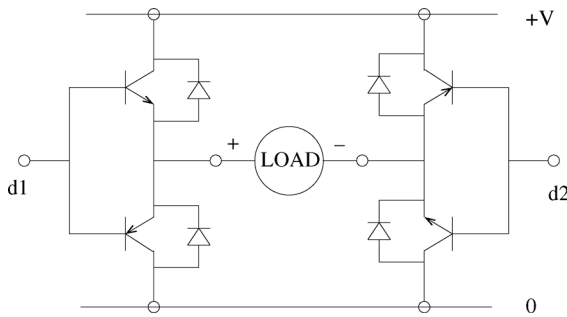


Fig. 4. Three-level PWM implementation (full-bridge).

TABLE I
TRUTH TABLE.

d1	d2	LOAD
0	0	0
V	0	V
0	V	-V
V	V	0

applications, ranging from power conversion to communications. However, it is well known that the process of PWM conversion is nonlinear and it is generally supposed that low-pass filter signal recovery is imperfect for PWM [5], [6]. The intrinsic distortion of many conventional PWM techniques has been investigated intensively in the literature [5]–[8]. There has also been considerable work in the area of optimal PWM [9]–[11], which aims to optimize some specific performance criteria, e.g., to minimize harmonics or eliminate specific harmonics in the PWM frequency spectrum. These works consider only periodic signals and their techniques are not applicable to the general signals with which this paper deals. The literature also proposes many distortion compensating techniques [12]–[20]. However, to the best of our knowledge, none of these distortion compensating techniques has achieved complete elimination of PWM distortions.

have this property, e.g., loudspeakers and electric motors; alternatively, a filter can be implemented explicitly using capacitors and inductors. The diodes are used to allow power to flow back from the load to the source. For the two-level PWM, there are two transitions (one up, one down) per clock cycle. For the three-level PWM, there is just one transition per push-pull stage per clock cycle. Therefore, it is possible to implement three-level PWM with lower frequency devices.

The technique of PWM has a long history of theoretical and practical development. Nowadays PWM is widely used in many

There do exist techniques in the literature which represent bandlimited signals exactly by pulsewidth modulated (binary) signals. Click modulation [21] and time encoding [24] are two such techniques. Logan in [21] develops click modulation and demonstrates that some bandlimited signals can be exactly represented by binary signals. The technique of click modulation and its several extensions [22], [23] investigate the same topic as the current paper. The technique first encodes the input signal $x(t)$ into $Z(t) = e^{-j(x(t)) + j\hat{x}(t)}$, where $\hat{x}(t)$ is the Hilbert transform of $x(t)$, and then generates $z(t)$ by low-pass filtering $Z(t)$.

Under the sufficient condition that $z(\tau)$ ($\tau = t + ju$) is zero-free

in the upper half-plane, this technique can generate a distortion free binary signal. Under the special case when $x(t)$ is a sinusoid whose fundamental frequency is half of the PWM switching frequency, this condition turns out to be the same as the peak constraint in our sampling theorem. However, for general signals, this condition is difficult to quantify precisely as stated in [21], while the peak constraint provided by this current paper is simple. Furthermore, our technique can always find the exact PWM representation whenever it exists. This is not the case for click modulation and its extensions. For example, our technique can support much higher magnitudes than $2/\pi$ for lower than PWM (switching) frequency sinusoids, while click modulation cannot. Our technique allows the input signal to have a dc component, while click modulation can only allow bandpass-type input signals. The hybrid ZePoC technique provided in [22] tries to solve this problem by treating the whole baseband as two separate bands. However, this technique is no longer distortion free. Time encoding [24] encodes the amplitude information of a baseband signal into the transition times of a binary signal. The time encoding mechanism generates irregular pulses, while the present paper considers regular pulses. In other words, the samples are irregularly spaced and the switching frequency of the time encoded waveform is not constant as in the present paper. It could be very large for certain input signals.

The paper begins by developing a sequential local error minimization technique which forms a convenient proof for the sampling theorem. This technique sequentially adjusts the pulsewidths to achieve the mapping between a bandlimited baseband signal and a PWM waveform. This technique can find the corresponding PWM waveform for any bandlimited baseband signal (even if it exceeds the peak constraint) provided there exists an exact PWM representation. The paper goes on to derive a different matrix based iterative technique which converges exponentially quickly when the peak is strictly less than $\frac{2}{\pi}$. The paper goes on to show that there are many practically interesting signals with higher peaks than the peak constraint, e.g., random signals or lower than PWM frequency sinusoids, that can be represented by PWM waveforms.

The paper organization is straightforward. Section II introduces the basic concepts of PWM and describes the problem setting. Section III presents and proves the sampling theorem with constant amplitude and variable width pulses. Section IV presents a matrix based iterative technique to generate the PWM waveform for a given baseband signal that satisfies the peak constraint in the sampling theorem. Section V shows that there are many practically interesting signals with higher peaks that can be represented by PWM waveforms. Section VI concludes this paper.

II. PROBLEM SETTING

The conventional Nyquist sampling theorem gives a mapping between samples and any bandlimited baseband signal. In classical sampling, the baseband signal is represented by a series of Dirac delta functions of varying amplitudes. PWM, on the other hand, represents a signal by using pulses of constant amplitude but variable widths. In that sense, PWM is a substitute for classical sampling.

This paper proves the sampling theorem with constant amplitude variable width pulses by mapping the baseband signals to a particular class of PWM waveforms. The input signal is uniformly time-sampled and the pulsewidth is calculated at each sampling time instant. The pulse is symmetrically distributed with respect to the sampling time instant. In other words, the centers of all pulses are fixed while two edges of each pulse are moved to change the width. The paper considers both two-level and three-level PWM, which are sometimes called class-AD and class-BD PWM respectively [7], [25]. The two-level PWM waveform has a voltage level of -1 and $+1$, while the three-level PWM waveform has a voltage level of -1 , 0 and $+1$.

Mathematically, the three-level PWM signal that we consider can be represented as a sum of pulses over all time, in terms of the Heaviside step function $u(t)$, as

$$pwm(t) = \sum_{n=-\infty}^{\infty} \left[u\left(t - nT + \frac{w_n T}{2}\right) - u\left(t - nT - \frac{w_n T}{2}\right) \right] \quad (2.1)$$

where $(n - w_n/2)T$ and $(n + w_n/2)T$ are the rising and falling edges of the n th pulse, $w_n T$ is the actual duration of the n th pulse, and T is the sampling period. We also define T as the PWM period and $1/T$ as the PWM (switching) frequency. The term w_n is constrained to be within the range of $[-1, 1]$ for three-level PWM. It is the normalized width of the n th pulse and we directly call it the width of the n th pulse in this paper. All pulses have unit magnitude. For two-level PWM, the term w_n is constrained to be within the range of $[0, 1]$. We can simply do the operation of $2 \cdot pwm(t) - 1$ to get the desired two-level PWM waveform. If $w_n > 0$, then the corresponding pulse is positive. If $w_n < 0$, then the corresponding pulse is negative. Thus, the two-level PWM signal consists of a sequence of only positive pulses, while the three-level PWM signal consists of a sequence of positive or negative pulses.

The goal of this paper is to represent a bandlimited baseband signal, $x(t)$, with a PWM waveform such that passing the PWM waveform through an ideal low-pass filter will yield back $x(t)$ without any distortion. We first perform a transformation on $x(t)$ so that we can use the same notation for both two-level and three-level PWM representations. We transform $x(t)$ to $s(t)$ as

$$s(t) = \begin{cases} \frac{x(t)+1}{2}, & \text{two-level} \\ x(t), & \text{three-level} \end{cases} \quad (2.2)$$

where two-level and three-level mean two-level PWM representation and three-level PWM representation respectively. For the two-level case, (2.2) transforms $x(t)$ to a positive only signal. The resulting two-level PWM waveform will consist of positive pulses. Of course, we can also proceed in the other direction by transforming $x(t)$ to a negative only signal, i.e., $s(t) = (x(t) - 1)/2$. The resulting two-level PWM waveform will consist of negative pulses. When the input signal $x(t)$ is not known, these two methods are equivalent. However, for particular instances of $x(t)$, one method may perform better than the other. Suppose the exact PWM waveform for $s(t)$ is $pwm(t)$. For three-level PWM representation, the PWM waveform for $x(t)$ is $pwm(t)$ ($w_n \in [-1, 1]$). For two-level PWM representation, the PWM waveform for $x(t)$ is $2 \cdot pwm(t) - 1$ ($w_n \in [0, 1]$).

The next few paragraphs describe the mapping between the PWM pulsewidths and the output of the low-pass filter. Suppose that we have a sequence of pulses with widths $\underline{w}$. Note that underlined letters are used to denote row vectors. We pass this sequence of pulses through a low-pass filter, we sample the output signal at the time instances of those pulses' center, and we get a sequence of samples $\hat{s}$. This filtering process can be represented as a function (mapping) between $\hat{s}$ and $\underline{w}$ as $\hat{s} = f(\underline{w})$. In the following, we explain this mapping in detail.

In the time domain, the output of low-pass filtering is given by the convolution [26] between the input and the impulse response of the filter, e.g., a sinc function. Let $f_n(w)$ denote the convolution between a unit magnitude pulse with width w and the portion of the normalized sinc function in the range of $[n - 0.5, n + 0.5]$. If $n = 0$, we refer to the main lobe. Otherwise, n is the zero crossing point of the normalized sinc function. The $f_n(w)$ function is given as

$$f_n(w) = \int_{n-w/2}^{n+w/2} \frac{\sin \pi t}{\pi t} dt \quad n = 0, 1, 2, \dots \quad (2.3)$$

The $f_n(w)$ function is continuous, nonlinear and monotonic. We also denote $f'_n(w)$ as the derivative function of $f_n(w)$ and $f_n^{-1}(w)$ as the inverse function of $f_n(w)$. For a given w , $f'_n(w)$ is effectively the sum of the values of the sinc function at $n - w/2$ and $n + w/2$.

Similar to the intersymbol interference (ISI) in digital communication, we use ISI to denote the interference to a particular pulse in the low-pass filtering process caused by all subsequent pulses. The ISI is the nonlinear interaction of all subsequent pulses with the present pulse. Mathematically, ISI is caused by the convolution between all the other pulses and their corresponding side lobes of the sinc function.

Definition 2.1: (Definition of ISI) The ISI to the t th pulse caused by the other pulses are given by

$$\text{ISI}_t = \sum_{n=-\infty, n \neq t}^{\infty} f_{|t-n|}(w_n). \quad (2.4)$$

Note that the ISI in this paper are calculated from a unit magnitude pulse and normalized pulsewidth.

The t th output sample $\hat{s}_t$ is the summation of the contribution from the t th pulse and the ISI due to all the other pulses as

$$\hat{s}_t = f_0(w_t) + \text{ISI}_t. \quad (2.5)$$

Therefore, given all the pulsewidths, it is easy to calculate the corresponding output samples since all the $f_n(\cdot)$ functions are deterministic and known. However, in the opposite direction, given all the samples, it is hard to calculate the corresponding pulsewidths. This is because of the nonlinearity in the $f_n(\cdot)$ functions and the fact that the PWM process is a nonlinear modulation. The following part of the paper will try to find the PWM representation of any given baseband signal.

III. THE SAMPLING THEOREM WITH VARIABLE WIDTH PULSES

This section proves the sampling theorem for constant amplitude and variable width pulses. The section first proves the result for three-level PWM and then uses this to demonstrate the result for two-level PWM. Subsection A derives a strict upper bound

on the magnitude of all the possible ISI to any single pulse. This upper bound is achieved if the PWM signal is a periodic square waveform. Subsection B uses a sequential local error minimization technique to prove this sampling theorem and then proves that even if the peak constraint is not satisfied, the sequential local error minimization technique will converge to the exact PWM representation whenever it exists.

A. The Worst ISI

This subsection derives a universal upper bound on the magnitude of any ISI to a single pulse. The upper bound is given in the following proposition.

Proposition 3.1: The periodic ideal square waveform causes the worst ISI. The worst ISI gives a strict upper bound ISI^m on the magnitude of the ISI that could happen. This upper bound is given by

$$\text{ISI}^m = 2 \sum_{n=1}^{\infty} \left(\int_{n-0.5}^n \left| \frac{\sin \pi t}{\pi t} \right| dt - \int_n^{n+0.5} \left| \frac{\sin \pi t}{\pi t} \right| dt \right), \quad (3.1)$$

where ISI^m can be evaluated in MATLAB and $\text{ISI}^m = 0.236$.

Proof: The magnitude of the ISI to the t th pulse can be bounded as

$$\begin{aligned} |\text{ISI}_t| &= \left| 2 \sum_{n=1}^{\infty} \left(\int_{n-w_n/2}^{n+w_n/2} \text{sign}(w_n) \cdot \frac{\sin \pi t}{\pi t} dt \right) \right| \\ &\stackrel{(a)}{\leq} 2 \sum_{n=1}^{\infty} \left(\int_{n-w_n/2}^{n+w_n/2} \left| \frac{\sin \pi t}{\pi t} \right| dt \right) \\ &= 2 \sum_{n=1}^{\infty} \left(\int_{n-|w_n|/2}^n \left| \frac{\sin \pi t}{\pi t} \right| dt - \int_n^{n+|w_n|/2} \left| \frac{\sin \pi t}{\pi t} \right| dt \right) \\ &\stackrel{(b)}{\leq} 2 \sum_{n=1}^{\infty} \left(\int_{n-0.5}^n \left| \frac{\sin \pi t}{\pi t} \right| dt - \int_n^{n+0.5} \left| \frac{\sin \pi t}{\pi t} \right| dt \right) \end{aligned} \quad (3.2)$$

where equality in (a) holds if the pulses in each side of the targeted pulse have alternating signs, e.g., $\text{sign}(w_n) \geq 0$ ($n = 2k$) and $\text{sign}(w_n) < 0$ ($n = 2k+1$) for $k = 1, 2, \dots$. Equality in (b) holds if $|w_n| = 1$. Therefore, we get a universal upper bound on the magnitude of the ISI for any particular single pulse. The upper bound is achieved when the PWM waveform is a periodic ideal square waveform. Clearly, the corresponding input signal is a sinusoid due to symmetry. The amplitude of the sinusoid is $f_0(1) - \text{ISI}^m = 0.637$. ■

B. Iterative Error Reduction

This subsection presents a sequential local error minimization technique to achieve the mapping between a bandlimited baseband signal and a PWM waveform. Fig. 5 presents a block diagram of the iterative scheme. Note that this subsection first focuses on the analysis of three-level PWM, and then extends the results to two-level PWM.

For any bandlimited signal $x(t)$ within bandwidth B , we generate $s(t)$ according to (2.2), and we sample $s(t)$ at the Nyquist rate ($1/T = 2B$, T : sampling period) in the time domain as in

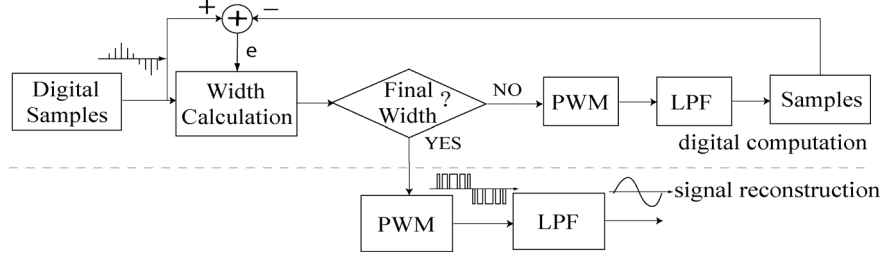


Fig. 5. Iterative error reduction diagram.

Fig. 2(a). Now suppose that we get a sequence of N samples $\underline{s}$, and we want to modulate it to a sequence of N pulses in a way such that if we pass the sequence of pulses into an ideal low-pass filter with cutoff frequency B , we will get back the original bandlimited signal without any distortion. Note that N is large enough that we can ignore edge effects.

The sequential local error minimization technique will calculate the widths for the final PWM waveform. This technique is an iterative process as shown above the dashed line in Fig. 5, and it is done purely in the digital domain. The technique starts with a conventional PWM technique. For the sequence of samples $\underline{s}$, we first generate the initial widths for the pulses which are centered at the time instances of the samples. Each pulsewidth is proportional to the magnitude of the corresponding sample

$$\underline{w} = \frac{1}{P} \cdot \underline{s}, \quad (3.3)$$

where P can be chosen to be the maximum absolute amplitude of the samples. The idea is illustrated in Fig. 2(b).

We can predict the output samples corresponding to $\underline{s}$ if we pass the PWM waveform with the calculated pulsewidths through a digital low-pass filter. We define the predicted output samples to be $\hat{\underline{s}}$. We then compare the output samples $\hat{\underline{s}}$ with the original input samples $\underline{s}$, and generate the error signal $\underline{e} = \underline{s} - \hat{\underline{s}}$. The error signal is then used as a feedback to adjust the widths of the pulses.

Definition 3.2: Let E denote the sum of the absolute values of the individual errors in $\underline{e}$, e.g.,

$$E = \sum_{n=1}^N |e_n|. \quad (3.4)$$

Since $|e_n| \geq 0$, if $E = 0$, then $\underline{e} = 0$.

Later we iteratively reduce the error to let the error $\underline{e}$ go to zero. In each iteration, we sequentially adjust the widths of the pulses. In other words, each iteration has N steps of width adjustment. More specifically, in the first step, we adjust the width of the first pulse such that its predicted output sample equals the corresponding original sample, i.e., $e_1 = 0$, with all other pulsewidths unchanged. We then repeat the same procedure for the n th pulsewidth when n increases one by one from 2 to N . We repeat the same sequential adjustment iteratively until the error $\underline{e}$ reduces to the desired level. In the following subsection, we prove that this iterative procedure makes the error term E go down in each iteration.

Proof of Convergence

This subsection proves that the procedure of iterative error reduction will bring the final output samples converge to the original input samples. In order to prove convergence, we prove that the sum of the absolute values of the individual errors in $\underline{e}$, i.e., E , goes down at any step of the sequential adjustment in each iteration if the change in the width is nonzero.

We first introduce a range of new parameters.

Definition 3.3: Let Δe_n denote the change of the absolute value of error e_n due to the change in the width of the targeted t th pulse. If Δe_n is positive, then $|e_n|$ is increased. If Δe_n is negative, then $|e_n|$ is reduced. Let ΔE_t denote the sum of the absolute values of the change in the other errors in $\underline{e}$, excluding e_t , that are caused by the change in the width of the targeted t th pulse, i.e.,

$$\Delta E_t = \sum_{n=1, n \neq t}^N |\Delta e_n|. \quad (3.5)$$

Since $\Delta E_t \geq \sum_{n=1, n \neq t}^N \Delta e_n$, ΔE_t must hold as an upper bound on the total error change in all the other individual samples caused by the change in the width of the targeted t th pulse.

Definition 3.4: Let $\tilde{E}$ denote the change in E due to the change in the width of the targeted t th pulse. If $\tilde{E} < 0$, then the total error E is reduced, otherwise E is increased or unchanged.

It is easy to show that $\tilde{E} = \Delta e_t + \sum_{n=1, n \neq t}^N \Delta e_n \leq \Delta e_t + \Delta E_t$. This is because ΔE_t holds as an upper bound on the total error change in all the other individual samples, $\Delta e_t + \Delta E_t$ must hold as an upper bound on $\tilde{E}$.

The proof of convergence involves two parts. The first part is that in any step of any iteration, we can always adjust the width of the targeted pulse such that the final corresponding output sample equals the original input sample. The second part is that each step of sequential adjustment decreases the total error in all samples at nonzero rate. The first part is given in the following proposition.

Proposition 3.5: In the t th step of any iteration, given all the other elements ($w_n \in [-1, 1]$, $n = 1, \dots, N$ and $n \neq t$) in $\underline{w}$ unchanged, we can always find a particular $w_t \in [-1, 1]$ such that the corresponding output sample $\hat{s}_t$ equals s_t , where s_t is any given sample that satisfies $s_t \in [-f_0(1) + \mathcal{IST}^m, f_0(1) - \mathcal{IST}^m]$.

Note that $\mathcal{IST}^m$ is achieved when the PWM waveform is an ideal square waveform. The corresponding output signal of low-pass filtering is a sinusoid. $f_0(1) - \mathcal{IST}^m$ is actually the amplitude of the sinusoid. Consider the convolution between an ideal square waveform with period 2 and unit magnitude and the

normalized sinc function. Using Fourier series, we can write this square waveform $sq(t)$ as an infinite series of this form

$$sq(t) = \frac{4}{\pi} \sum_{k=1}^{+\infty} \frac{\sin((2k-1)\pi t)}{2k-1}. \quad (3.6)$$

The convolution between $sq(t)$ and the normalized sinc function gives $(2/\pi)\sin(\pi t)$. Thus, we have $f_0(1) - \mathcal{ISI}^m = (2/\pi) = 0.637$.

Proof: The proof of Proposition 3.5 is straightforward. Equation (2.5) gives the relationship between $\hat{s}_t$ and w_t . Since $f_0(w)$ is a continuous function of w and $f_0(w) \leq f_0(1)$, if $|s_t - \mathcal{ISI}^m| \leq f_0(1)$, then we can always find a $w_t \in [-1, 1]$ corresponding to s_t . From Proposition 3.1, we know that $\mathcal{ISI}_t$ is constrained as $|\mathcal{ISI}_t| \leq \mathcal{ISI}^m$. Since $s_t \in [-f_0(1) + \mathcal{ISI}^m, f_0(1) - \mathcal{ISI}^m]$, then it is easy to show that $|s_t - \mathcal{ISI}^m| \leq f_0(1)$. Thus, this proposition is proved. ■

The following proposition gives the second part of the proof of convergence.

Proposition 3.6: In the t th step of any iteration, the procedure of adjusting the width of t th pulse such that $\hat{s}_t = s_t$ with all other pulses unchanged, $\tilde{E}$ will always be negative. In other words, each step will let the total error E of all the samples go down at nonzero rate when the change in the width is nonzero.

Proof: We have shown that $\tilde{E} \leq \Delta e_t + \Delta E_t$. Without loss of generality, we assume $w_t \geq 0$ and $\Delta w > 0$. To prove this proposition, we first get an upper bound on ΔE_t

$$\begin{aligned} \Delta E_t &= 2 \sum_{n=1}^{N-1} \left(\int_{n-w_t/2-\Delta w}^{n-w_t/2} \left| \frac{\sin \pi t}{\pi t} \right| dt - \int_{n+w_t/2}^{n+w_t/2+\Delta w} \left| \frac{\sin \pi t}{\pi t} \right| dt \right) \\ &\stackrel{(a)}{<} 2 \int_{1-w_t/2-\Delta w}^{1-w_t/2} \left| \frac{\sin \pi t}{\pi t} \right| dt, \end{aligned} \quad (3.7)$$

where (a) holds since $\int_{n+w_t/2}^{n+w_t/2+\Delta w} |\sin \pi t / \pi t| dt \geq \int_{n+1-w_t/2-\Delta w}^{n+1-w_t/2} |\sin \pi t / \pi t| dt$.

We then show that $\Delta e_t + \Delta E_t < 0$, where Δe_t is given by

$$\Delta e_t = -2 \int_{w_t/2}^{w_t/2+\Delta w} \left| \frac{\sin \pi t}{\pi t} \right| dt. \quad (3.8)$$

Since $0 < w_t/2 + \Delta w \leq 1 - w_t/2 - \Delta w < 1$, it is obvious that $|\Delta e_t|$ is larger than $2 \int_{1-w_t/2-\Delta w}^{1-w_t/2} |\sin \pi t / \pi t| dt$, which is the upper bound on ΔE_t we derived before. Thus, we have

$$\tilde{E} \leq \Delta e_t + \Delta E_t < 0. \quad (3.9)$$

For three-level PWM representation, we have the following theorem.

Theorem 3.7: (Three-level PWM) Any bandlimited signal $x(t)$ within bandwidth B and $|x(t)| \leq 2/\pi = 0.637$, can be represented by a three-level PWM waveform with frequency $2B$ and voltages 0 and ± 1 . Low-pass filtering can recover $x(t)$ perfectly from the PWM waveform.

Proof: Proposition 3.5 and 3.6 can be combined to show that the iterations converge to a sequence of widths such that the error between the filtered PWM waveform and the original

input samples goes to zero. Thus, this iterative procedure forms a constructive proof for the sampling theorem with three-level waveforms. ■

The next theorem proves that the results can easily be extended to the two-level case. The counterpart of Theorem 3.7 for two-level PWM can be stated as follows.

Theorem 3.8: (Two-level PWM) Any bandlimited signal $x(t)$ within bandwidth B and $|x(t)| \leq 2/\pi = 0.637$, can be represented by a two-level PWM waveform with frequency $2B$ and voltages ± 1 . Low-pass filtering can recover $x(t)$ perfectly from the PWM waveform.

Proof: Consider the PWM waveform given by (2.1) when w_n is allowed to be in $[0, 1]$. Note that this is a PWM waveform with two levels 0 and 1 . We can bound the ISI to the t th pulse as

$$2 \sum_{n=2,4,6,\dots} f_n(1) \leq \mathcal{ISI}_t \leq 2 \sum_{n=1,3,5,\dots} f_n(1), \quad (3.10)$$

where the lower and upper bounds are denoted as $\mathcal{ISI}^{mL}$ and $\mathcal{ISI}^{mU}$. We have $\mathcal{ISI}^{mU} + \mathcal{ISI}^{mL} = 1 - f_0(1)$ and $\mathcal{ISI}^{mU} - \mathcal{ISI}^{mL} = \mathcal{ISI}^m$. The above iterative error reduction method for three-level PWM can also be applied to two-level (0 and 1) PWM to prove the similar results. The counterpart of Proposition 3.5 for two-level (0 and 1) PWM can be stated as: In the t th step of any iteration, given all the other elements ($w_n \in [0, 1]$, $n = 1, \dots, N$ and $n \neq t$) in $\underline{w}$ unchanged, we can always find a particular $w_t \in [0, 1]$ such that the corresponding output sample $\hat{s}_t$ equals s_t , where s_t is any given sample that satisfies $s_t \in [\mathcal{ISI}^{mU}, f_0(1) + \mathcal{ISI}^{mL}]$.

The counterpart of Proposition 3.6 is the same for two-level PWM. Therefore, a PWM waveform with two levels 0 and 1 can represent a baseband signal in the range of $[\mathcal{ISI}^{mU}, f_0(1) + \mathcal{ISI}^{mL}]$ exactly. By doing the operation of $2 \cdot pwm(t) - 1$, we can transfer the PWM waveform with two levels 0 and 1 to the desired PWM waveform with levels -1 and 1 . Since low-pass recovery is a linear operation, we can conclude that two-level PWM waveform can represent a baseband signal within the range of $[2\mathcal{ISI}^{mU} - 1, 2f_0(1) + 2\mathcal{ISI}^{mL} - 1]$. It is easy to show that $2\mathcal{ISI}^{mU} - 1 = -(f_0(1) - \mathcal{ISI}^m) = -2/\pi$ and $2f_0(1) + 2\mathcal{ISI}^{mL} - 1 = 2/\pi$. This proves the theorem. ■

Even though it is generally supposed that low-pass recovery of PWM waveforms results in distortion in the modulating signal [5], [6], both Theorem 3.7 and Theorem 3.8 tell us that PWM representation can be perfect. Note that it makes no difference whether the PWM waveform is two-level or three-level with respect to the peak constraints in both theorems. The number of pulses for this PWM representation is the same as the number of Nyquist samples. Simple as it is, the peak constraint given in both theorems is not found in the literature.

The two theorems provide a sufficient condition on the peak constraint for exact PWM representation. For either two-level PWM or three-level PWM, when the input signal satisfies the peak constraint in the corresponding theorem, the pulsewidths will always be in the range of $[0, 1]$ (for two-level PWM) or $[-1, 1]$ (for three-level PWM) in the sequential local error minimization technique. In terms of the peak constraint, the constraint in the two theorems is tight for the sufficient condition.

This is because we can always find baseband signals which

cannot be represented by either two-level PWM or three-level PWM exactly if the peak constraint is violated. For example, if the amplitude for a sinusoid whose fundamental frequency is half of the PWM switching frequency is larger than the peak constraint, then it is impossible to represent it exactly by either a two-level PWM waveform or a three-level PWM waveform. The peak constraint given by the two theorems is not a necessary condition, and some signals with higher peaks can also be represented by PWM waveforms exactly. Specific examples will be given in Section V. For signals with higher peaks, in the sequential local error minimization technique it is possible that some pulses are already at the full width, however, there are still errors with these pulses, e.g., the corresponding output samples are not the same as the input samples. We call these pulses saturated pulses. In the following we will show that whenever there exists an exact PWM representation, the iterative technique will converge to the correct answer. This is valid for both two-level PWM and three-level PWM. The proof involves the following proposition.

Proposition 3.9: Consider the case when the sequential local error minimization technique converges, but the total error E is not zero. Then the errors only exist at those saturated pulses, while the pulses that are not saturated are free of error.

Proof: This proposition can be proved by contradiction. Assume when the sequential local error minimization technique converges, there exist some pulses which are not saturated and have errors. According to Proposition 3.6, we can always adjust the widths of those particular pulses one by one to decrease the total error E . This contradicts with the assumption that the sequential local error minimization technique has converged. ■

Theorem 3.10: When there exists an exact PWM representation of a bandlimited baseband signal, the sequential local error minimization technique will always converge to the correct answer.

Proof: This theorem can also be proved by contradiction. Assume that there exists an exact PWM representation of the bandlimited baseband input signal, and when the sequential local error minimization technique converges, there exist some saturated pulses with errors. For those saturated pulses with errors, we cannot increase their pulsewidths to remove the errors. The only way we can do is to change other pulses to adjust the ISI to those saturated pulses with errors. However, from Proposition 3.6, any change in any other pulse width will increase the total error E . Therefore the sequential local minimization technique converges to a local minimum. Furthermore, in any given direction, there is a unique minimum for the total error E . Therefore, there cannot be multiple local minima since the line joining any two will have multiple minima. Therefore, the iterative technique converges to a global minimum in the total error E . Hence, there is no exact PWM representation of the input signal. This contradicts with the assumption. Therefore, the theorem is proved. ■

IV. MATRIX BASED ITERATIVE TECHNIQUE

The previous section proved the sampling theorem which states that any bandlimited signal within a peak constraint can

be perfectly represented by a PWM waveform. The pulsewidths in the PWM waveform are allowed to be within the range of $[0, 1]$ for two-level PWM or $[-1, 1]$ for three-level PWM. Now suppose each pulsewidth in the PWM waveform is constrained to satisfy $0 \leq w \leq \tilde{w}$ for two-level PWM or $|w| \leq \tilde{w}$ for three-level PWM ($0 < \tilde{w} \leq 1$). We call this waveform the constrained PWM waveform. This section presents a matrix based iterative technique to generate the PWM waveform for a given peak constrained bandlimited signal $x(t)$. This matrix based iterative technique is performed purely in the digital domain and it converges exponentially with the number of iterations. This section also shows that small reduction in $\tilde{w}$ leads to significantly increased convergence rate.

We first show that the peak constraint in the sampling theorem for three-level PWM can actually be written as a function of the maximum allowed pulsewidth $\tilde{w}$ in the PWM waveform. Denote the peak constraint as $P(\tilde{w})$, e.g., $|x(t)| < P(\tilde{w})$. Following the proof for Proposition 3.5, we can easily write $P(\tilde{w})$ as a function of $\tilde{w}$ as

$$P(\tilde{w}) = f_0(\tilde{w}) - \mathcal{IST}^m(\tilde{w}), \quad (4.1)$$

where $\mathcal{IST}^m(\tilde{w})$ is defined as

$$\mathcal{IST}^m(\tilde{w}) = 2 \sum_{n=1}^{\infty} \left(\int_{n-\tilde{w}/2}^n \left| \frac{\sin \pi t}{\pi t} \right| dt - \int_n^{n+\tilde{w}/2} \left| \frac{\sin \pi t}{\pi t} \right| dt \right). \quad (4.2)$$

The derivation of $\mathcal{IST}^m(\tilde{w})$ follows similarly as the derivation of $\mathcal{IST}^m$ in the proof for Proposition 3.1. Equation (4.1) holds mainly because of the monotonicity in the $f_n(\cdot)$ functions. Then Theorem 3.8 can be rephrased as: Any bandlimited signal $x(t)$ within bandwidth B and $|x(t)| < P(\tilde{w})$, can be represented by a constrained PWM waveform ($0 \leq w \leq \tilde{w}$) with frequency $2B$ and voltages ± 1 .

The matrix based iterative technique is as follows. As in the previous section, we generate $s(t)$ according to (2.2). After getting the Nyquist samples $\underline{s}$, we also start with a conventional PWM technique defined in (3.3), e.g., $\underline{w}^1 = (1/P(\tilde{w}))\underline{s}$, to calculate the initial pulsewidths. Following the same idea as in Fig. 5, we then iteratively reduce the error to the desired level. At the i th iteration, we can write the t th element in the output samples $\hat{\underline{s}}^i$ in terms of the pulsewidths $\underline{w}^i$ as

$$\hat{s}_t^i = f_0(w_t^i) + \sum_{n=1, n \neq t}^N f_{|t-n|}(w_n^i), \quad (4.3)$$

where the superscripts are used to denote the number of iteration. The technique compares the output samples $\hat{\underline{s}}^i$ with the original input samples $\underline{s}$, and generates the error signal $\underline{e}^i$ as

$$\underline{e}^i = \underline{s} - \hat{\underline{s}}^i. \quad (4.4)$$

The error signal is then used to adjust the widths of the pulses as

$$\underline{w}^{i+1} = \underline{w}^i + \underline{e}^i \mathbf{A}^{-1} \quad (4.5)$$

where $\mathbf{A}$ is a constant $N \times N$ matrix given in the form as

$$\mathbf{A} = \begin{pmatrix} f'_0(w) & f'_1(w) & \cdots & f'_{N-1}(w) \\ f'_1(w) & f'_0(w) & \cdots & f'_{N-2}(w) \\ \vdots & \vdots & \ddots & \vdots \\ f'_{N-1}(w) & f'_{N-2}(w) & \cdots & f'_0(w) \end{pmatrix}. \quad (4.6)$$

Note that $\mathbf{A}$ is a symmetric Toeplitz matrix. The reason for matrix $\mathbf{A}$ to be in this form is discussed as follows. Consider the case when the change in the widths is extremely small such that linear approximation can be considered to be accurate. We write the output samples at the $(i+1)$ th iteration as

$$\hat{\underline{s}}^{i+1} = f(\underline{w}^{i+1}) = f(\underline{w}^i + \underline{e}^i \mathbf{A}^{-1}) \quad (4.7)$$

and we write the error signal as

$$\begin{aligned} \underline{e}^{i+1} &= \underline{s} - \hat{\underline{s}}^{i+1} = \underline{s} - f(\underline{w}^i + \underline{e}^i \mathbf{A}^{-1}) \\ &\stackrel{(a)}{=} \underline{s} - \hat{\underline{s}}^i - \underline{e}^i \mathbf{A}^{-1} \mathbf{B} = \underline{e}^i [\mathbf{I} - \mathbf{A}^{-1} \mathbf{B}] \end{aligned} \quad (4.8)$$

where $\mathbf{I}$ stands for the identity matrix, and (a) follows from the assumption that linear approximation is accurate, and $\mathbf{B}$ is given by

$$\mathbf{B} = \begin{pmatrix} f'_0(w_1^i) & f'_1(w_1^i) & \cdots & f'_{N-1}(w_1^i) \\ f'_1(w_2^i) & f'_0(w_2^i) & \cdots & f'_{N-2}(w_2^i) \\ \vdots & \vdots & \ddots & \vdots \\ f'_{N-1}(w_N^i) & f'_{N-2}(w_N^i) & \cdots & f'_0(w_N^i) \end{pmatrix}. \quad (4.9)$$

Clearly, the matrix $\mathbf{A}$ should be close to $\mathbf{B}$ such that the eigenvalues of $\mathbf{I} - \mathbf{A}^{-1} \mathbf{B}$ have small magnitude. However, the widths in $\mathbf{B}$ vary in different iterations and for different input signals. From the computational complexity point of view, it is better for $\mathbf{A}$ to be a constant matrix such that its inverse can be precalculated and stored for use. Thus, we choose $\mathbf{A}$ to be in the form as (4.6). Note that the width w for $\mathbf{A}$ cannot be 1 because if $w = 1$, then $\mathbf{A}$ is singular. For other w , $\mathbf{A}$ will always be nonsingular. Furthermore, for convenient analysis, we only consider the matrix $\mathbf{A}$ to be the special case when the pulsewidth w is zero, i.e., $f'_0(0) = 2$ and $f'_n(0) = 0$ ($n \neq 0$). The matrix $\mathbf{A}$ is a diagonal matrix with all diagonal term to be $f'_0(0)$. Let $\Delta \underline{w}^i$ denote the change in the pulsewidths in the i th iteration, i.e., $\Delta \underline{w}^i = \underline{e}^i \mathbf{A}^{-1}$. Then the change in the t th pulsewidth is $\Delta w_t^i = (1/f'_0(0))e_t^i$. Without loss of generality, assume $\Delta w_t^i \neq 0$. At the $(i+1)$ th iteration, the new t th output sample becomes

$$\hat{s}_t^{i+1} = f_0(w_t^i + \Delta w_t^i) + \sum_{n=1, n \neq t}^N f_{|t-n|}(w_n^i + \Delta w_n^i), \quad (4.10)$$

and the new t th error becomes $e_t^{i+1} = s_t - \hat{s}_t^{i+1}$, whose amplitude can be written as

$$\begin{aligned} |e_t^{i+1}| &= |e_t^i - (\hat{s}_t^{i+1} - \hat{s}_t^i)| \\ &= |e_t^i - (f_0(w_t^i + \Delta w_t^i) - f_0(w_t^i)) \\ &\quad + \sum_{n=1, n \neq t}^N (f_{|t-n|}(w_n^i + \Delta w_n^i) - f_{|t-n|}(w_n^i))|. \end{aligned} \quad (4.11)$$

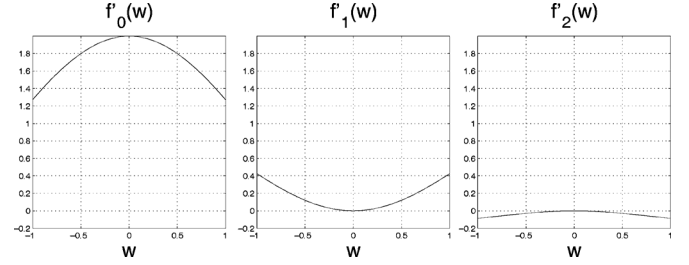


Fig. 6. Illustration of the $f'_i(w)$ functions.

In the following, we bound $|e_t^{i+1}|$ and show that this iterative technique converges exponentially. Fig. 6 illustrates the derivative functions $f'_n(w)$ for $n = 0, 1, 2, \dots$. As shown in Fig. 6, $f'_n(w)$ is symmetric with respect to $w = 0$, and monotonic for $w \geq 0$. According to mean value theorem, we have

$$f'_0(\tilde{w}) \leq \frac{f_0(w_t^i + \Delta w_t^i) - f_0(w_t^i)}{\Delta w_t^i} \leq f'_0(0). \quad (4.12)$$

By simple manipulation, we have

$$\frac{e_t^i}{\Delta w_t^i} - \frac{f_0(w_t^i + \Delta w_t^i) - f_0(w_t^i)}{\Delta w_t^i} \leq \frac{e_t^i}{\Delta w_t^i} - f'_0(\tilde{w}). \quad (4.13)$$

Since $f'_0(0) > f'_0(\tilde{w})$ and $e_t^i/\Delta w_t^i = f'_0(0)$, we have

$$|e_t^i - (f_0(w_t^i + \Delta w_t^i) - f_0(w_t^i))| \leq \frac{f'_0(0) - f'_0(\tilde{w})}{f'_0(0)} |e_t^i|. \quad (4.14)$$

Similarly, we have

$$|(f_{|n-t|}(w_n^i + \Delta w_n^i) - f_{|n-t|}(w_n^i))| \leq \frac{|f'_{|n-t|}(\tilde{w})|}{f'_0(0)} |e_n^i| \quad (4.15)$$

for $n = 1, 2, \dots, N$ and $n \neq t$.

Let γ^i denote the maximum absolute value of the elements of $\underline{e}^i$, i.e., $|e_t^i| \leq \gamma^i$ for $t = 1, 2, \dots, N$. Combining (4.11), (4.14) and (4.15), we can bound the amplitude of e_t^{i+1} as

$$|e_t^{i+1}| \leq k(\tilde{w}) \gamma^i, \quad (4.16)$$

where

$$k(\tilde{w}) = \frac{f'_0(0) - f'_0(\tilde{w}) + 2 \sum_{j=1}^{\infty} |f'_j(\tilde{w})|}{f'_0(0)}. \quad (4.17)$$

Let γ^{i+1} denote the maximum absolute value of the elements of $\underline{e}^{i+1}$, we then have

$$\gamma^{i+1} \leq k(\tilde{w}) \gamma^i. \quad (4.18)$$

Here $k(\tilde{w})$ represents the worst case convergence rate. Equality in (4.16) holds only when all elements in $\underline{e}^i$ have magnitude γ^i and have alternating signs, e.g., $\underline{e}^i = [\cdots \gamma^i - \gamma^i \gamma^i - \gamma^i \cdots]$. Generally speaking, this iterative technique should converge much faster than this rate. Combining (4.1) and (4.17), we can get $k(\tilde{w})$ as a function of the input signal peak constraint $P(\tilde{w})$ as illustrated in Fig. 7. This figure shows that with a little sacrifice in the peak constraint, we can get much better convergence rate.

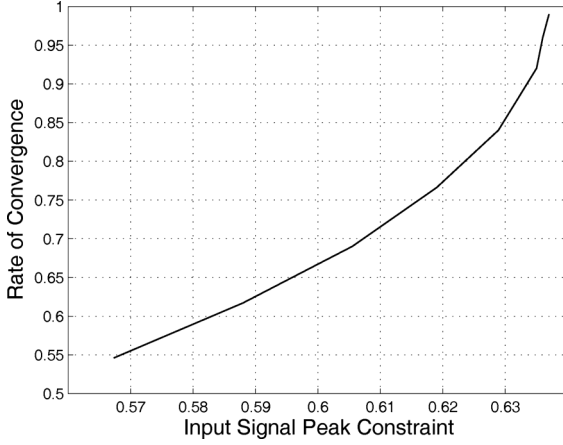


Fig. 7. The worst case rate of convergence.

V. EXTENSIONS TO SIGNALS WITH HIGHER PEAKS

The sampling theorem presented in Section III only provides a sufficient condition for the peak constraint of 0.637, but not a necessary one. There are many practically interesting signals with higher peaks that can be represented by PWM waveforms, for example, lower than PWM frequency sinusoids. With most type of random signals, there is no real peak penalty associated with PWM representation. For example, consider bandlimited Gaussian signals with a given power. It is impossible to represent them exactly by peak constrained samples with peak equal to 1 using the classic Nyquist sampling theorem. Some samples will be above the peak constraint, thus, there will be a clipping probability which is a function of the signal power. We will show in the following that there will be a similar clipping probability if they are represented by PWM waveforms. Obviously, this section cannot deal with all different types of signals, so it restricts itself to two special cases. Bandlimited Gaussian signals are discussed in Subsection A and low frequency sinusoids are discussed in Subsection B.

A. Bandlimited Gaussian Signals

This subsection considers the PWM representation of bandlimited white Gaussian signals. Gaussian signals are of significant importance and are widely considered in engineering systems. Most engineering applications are peak-limited while many random signals are not. Thus the random signals have to be clipped to satisfy the peak constraints. For such a scenario, it is well known that Nyquist samples can represent a bandlimited white Gaussian signal of power σ_N^2 with a certain clipping probability for a given peak constraint. The clipping probability is an important parameter when designing such systems. It is a measure of the distortion introduced by the peak constraint. Analogous to peak constrained Nyquist samples, the PWM representation is constrained by the maximum width of each pulse. Hence, we can calculate a similar clipping probability for the PWM representation. Suppose Nyquist samples with a peak constraint 1 can represent a bandlimited white Gaussian signal of power σ_N^2 for a given clipping probability. With the same clipping probability, a PWM waveform can represent a bandlimited white Gaussian signal of power σ_P^2 . This subsection will compare σ_N^2 with σ_P^2 , and it will focus on the three-level PWM representation to present the results. Each width of the PWM waveform is

allowed to be in the range of $[-1, 1]$. However, the results can be easily extended to two-level PWM case.

Consider a zero mean white Gaussian signal of power σ^2 whose frequency band is limited in the range of $[-B, B]$. We uniformly sample it at the Nyquist rate ($1/T = 2B$, T : sampling period) in the time domain and we get the samples $\underline{h}$. Note that all the elements in $\underline{h}$ will be zero mean and independent and identically distributed (i.i.d.) Gaussian random variables. Consider any single element h_t in $\underline{h}$. The probability density function (pdf) of h_t can be written as

$$p_{h_t}(h_t) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-h_t^2/2\sigma^2}, \quad (5.1)$$

where σ^2 is the variance of h_t .

Suppose the samples $\underline{h}$ are passed through a clipping device, and the clipping function $\lambda(h)$ is defined as

$$\lambda(h) = \begin{cases} h, & \text{if } |h| \leq h_0 \\ h_0 \operatorname{sign}(h), & \text{if } |h| > h_0 \end{cases},$$

where h_0 is the peak constraint. This paper assumes $h_0 = 1$ to be equal to the pulsewidth constraint. We name this procedure as natural Gaussian clipping. The clipping probability is the probability that the magnitude of h_t , e.g., $|h_t|$, is larger than h_0 , and it can be calculated as

$$\rho = \Pr[|h_t| > h_0] = 1 - \operatorname{erf}\left(\frac{h_0}{\sqrt{2\sigma^2}}\right), \quad (5.2)$$

where $\operatorname{erf}(\cdot)$ is the error function.

Now suppose we map the Gaussian samples $\underline{h}$ into a PWM waveform with the t th pulsewidth following the mapping rule

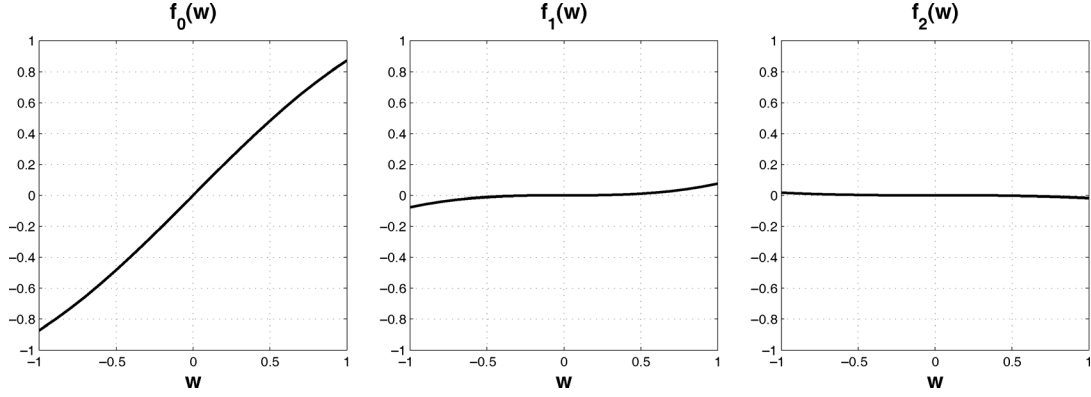
$$w_t = \begin{cases} 1, & \text{if } h_t - \mathcal{ISI}_t > f_0(1) \\ f_0^{-1}(h_t - \mathcal{ISI}_t), & \text{if } |h_t - \mathcal{ISI}_t| \leq f_0(1) \\ -1, & \text{if } h_t - \mathcal{ISI}_t < -f_0(1) \end{cases}.$$

Similarly, we name this procedure as PWM clipping. It is clear that the random variable $h_t - \mathcal{ISI}_t$ has zero mean and symmetric distribution. Since the inverse function $f_0^{-1}(\cdot)$ is a monotonic increasing odd function, we can write the PWM clipping probability ρ_{PWM} as

$$\rho_{\text{PWM}} = 2 \cdot \Pr[h_t - \mathcal{ISI}_t > f_0(1)]. \quad (5.3)$$

For a bandlimited white Gaussian signal, it is easy to calculate the natural Gaussian clipping probability following (5.2). However, this is not the case for the PWM clipping probability. This is because we do not have a closed form expression for the term $\mathcal{ISI}_t$ in (5.3). Actually, $\mathcal{ISI}_t$ cannot be calculated exactly without an iterative procedure. In the following we will evaluate the PWM clipping probability ρ_{PWM} by bounding $\mathcal{ISI}_t$. Consider the $\mathcal{ISI}_t$ term in equation (5.3). Following the definition in (2.4), we can write $\mathcal{ISI}_t$ as

$$\begin{aligned} \mathcal{ISI}_t &= \sum_{n=-\infty, n \neq t}^{+\infty} f_{|n-t|}(w_n) \\ &= \sum_{n=-\infty, n \neq t}^{+\infty} f_{|n-t|}(f_0^{-1}(h_n - \mathcal{ISI}_n)). \end{aligned} \quad (5.4)$$


 Fig. 8. Illustration of the $f_i(w)$ functions.

Similar to ISI_t , ISI_n is the ISI to h_n . Clearly, there is dependence between ISI_n and h_t . It is difficult to quantify this dependence, and thus difficult to evaluate the PWM clipping probability ρ_{PWM} directly. In order to avoid this dependence, we will not calculate ρ_{PWM} exactly. Instead we derive an upper bound, which involves only independent terms, on $h_t - ISI_t$. We then calculate an upper bound on ρ_{PWM} .

In Proposition 3.1, we derived a universal bound on the magnitude of ISI to any sample, i.e., $|ISI_t| \leq ISI^m$. In order to better evaluate the PWM clipping probability, we will derive tighter lower bounds on ISI_t for any particular realization of $\underline{h}$ in the following. Consider the calculation of ISI_t . Suppose we only consider the actual realization of those samples in a window of size $2T$, i.e., $[h_{t-T} \cdots h_{t-1} \ h_{t+1} \cdots h_{t+T}]$. The other samples outside the window are treated to maximize their contribution to ISI_t as

$$\begin{aligned} ISI_t &\stackrel{(a)}{\geq} \sum_{n=t-T, n \neq t}^{t+T} f_{|n-t|}(f_0^{-1}(h_n - ISI_n)) - ISI^T \\ &\stackrel{(b)}{\geq} \sum_{n=t-T, n \neq t}^{t+T} f_{|n-t|}(f_0^{-1}(h_n + (-1)^{|n|} \cdot ISI^m)) \\ &\quad - ISI^T, \end{aligned} \quad (5.5)$$

where $ISI^T = 2 \sum_{n=T+1}^{+\infty} |f_n(1)|$, and (a) in equation (5.5) holds since $f_n(\cdot)$ ($n \geq 0$) is a monotonic odd function, i.e., $|f_n(\cdot)| \leq |f_n(1)|$. (b) in equation (5.5) holds because the magnitude of ISI_n is upper bounded by ISI^m , and the $f_n(\cdot)$ is a monotonic increasing odd function if n is odd, and is a monotonic decreasing odd function if n is even. Fig. 8 illustrates $f_0(\cdot)$, $f_1(\cdot)$ and $f_2(\cdot)$. The righthand side of (b) involves only $2T$ independent Gaussian random variables in the window $[h_{t-T} \cdots h_{t-1} \ h_{t+1} \cdots h_{t+T}]$. Denote the righthand side of (b) in equation (5.5) as $ISI^{L,1}$. This lower bound does not involve ISI_n which is difficult to be evaluated exactly. This lower bound can be easily and exactly calculated for any particular realization of $[h_{t-T} \cdots h_{t-1} \ h_{t+1} \cdots h_{t+T}]$.

Instead of using $(-1)^{|n|+1} \cdot ISI^m$ to bound ISI_n as in (b) of equation (5.5), we can derive tighter lower or upper bounds

on ISI_n . We can therefore calculate a tighter lower bound on ISI_t as

$$ISI_t \geq \sum_{n=t-T, n \neq t}^{t+T} f_{|n-t|}(f_0^{-1}(h_n - ISI_n^b)) - ISI^T, \quad (5.6)$$

where ISI_n^b is either upper or lower bound on ISI_n if $|n-t|$ is odd or even correspondingly, and ISI_n^b is given by

$$\begin{aligned} ISI_n^b &= \sum_{j=t-T, j \neq n}^{t+T} f_{|j-n|}(f_0^{-1}(h_j + (-1)^{|j-n|+|n-t|} \cdot ISI^m)) \\ &\quad + (-1)^{|n-t|+1} \left(\sum_{j=\alpha 1}^{+\infty} |f_j(1)| + \sum_{j=\alpha 2}^{+\infty} |f_j(1)| \right), \end{aligned} \quad (5.7)$$

where $\alpha 1 = |t-T-n|+1$ and $\alpha 2 = |t+T-n|+1$. Denote the righthand side of equation (5.6) as $ISI^{L,2}$. $ISI^{L,2}$ is a tighter lower bound on ISI_t than $ISI^{L,1}$. However, $ISI^{L,2}$ is more difficult to calculate than $ISI^{L,1}$.

Therefore, we can easily upper bound $h_t - ISI_t$ as follows:

$$h_t - ISI_t \leq \hat{h}_T, \quad (5.8)$$

where

$$\hat{h}_T = h_t - ISI^{L,k}, \quad (5.9)$$

with k to be either 1 or 2.

Thus, the PWM clipping probability in (5.3) can be upper bounded as

$$\rho_{\text{PWM}} \leq 2 \cdot \Pr[\hat{h}_T > f_0(1)], \quad (5.10)$$

where the probability term in the right-hand side can be evaluated as

$$\begin{aligned} \Pr[\hat{h}_T > f_0(1)] &= \underbrace{\int \cdots \int}_{\hat{h}_T > f_0(1)} p_{h_t}(h_t) p_{h_{t-T}}(h_{t-T}) \cdots p_{h_{t+T}}(h_{t+T}) dh_t dh_{t-T} \cdots dh_{t+T}. \end{aligned} \quad (5.11)$$

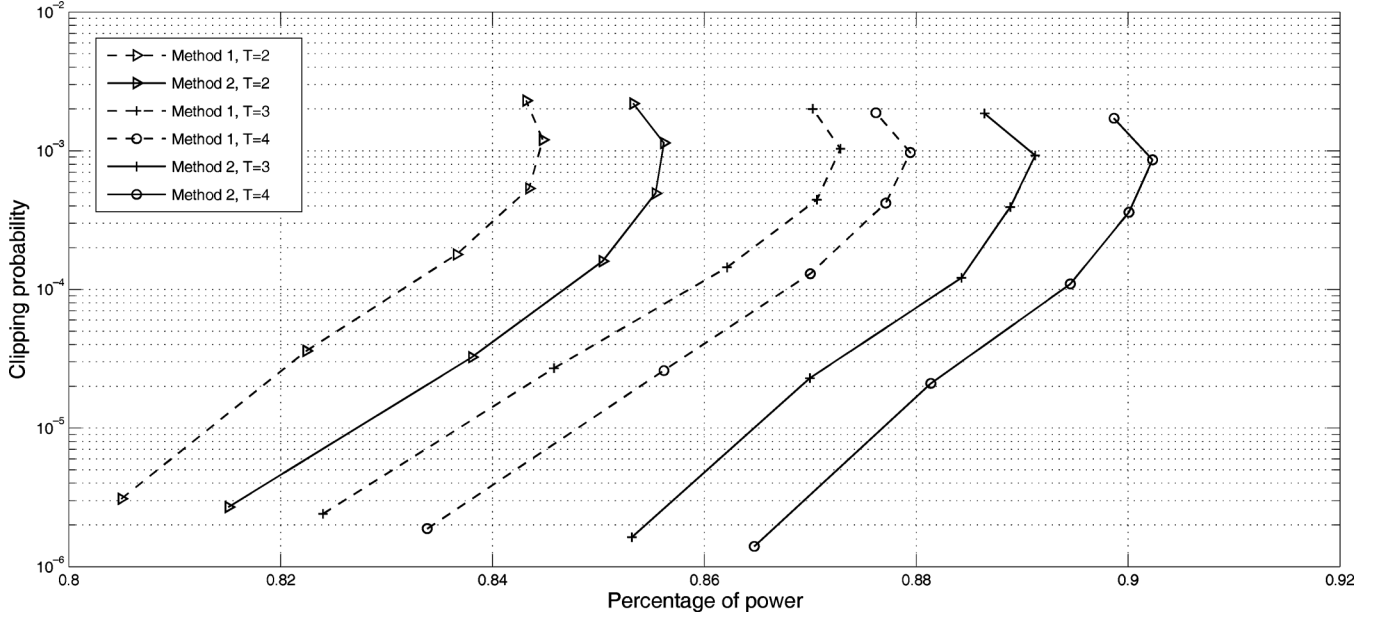


Fig. 9. Performance of PWM representation of bandlimited white Gaussian signals.

The $2T + 1$ integrals in (5.11) can be evaluated in MATLAB. As T increases, the upper bound in (5.10) will become closer to the actual PWM clipping probability. However, without any optimization, it is apparent that the computational complexity of evaluating the integrals in (5.11) goes exponentially with T . Some optimization techniques can be applied to reduce the complexity significantly. The side lobes of the sinc function decay exponentially, so do the $f_n(\cdot)$ functions. In other words, as n increases, the contribution of $f_n(\cdot)$ to the corresponding calculation of ISI decays exponentially. Therefore, small window sizes can give comparatively good results.

Consider the term σ_P^2/σ_N^2 . For PWM representation of general zero mean random signals, the term σ_P^2/σ_N^2 is less than $f_0(1)^2 = 0.7615$. This is because for any particular sample h_t , the corresponding ISI $\mathcal{ISI}_t$ is zero mean and has dependence on h_t . Combining the fact that $f_0(\cdot) \leq f_0(1)$ and $h_t = f_0(w_t) + \mathcal{ISI}_t$, we have

$$\Pr[|h_t - \mathcal{ISI}_t| > f_0(1)] > \Pr[|h_t| > f_0(1)] = \Pr\left[\left|\frac{h_t}{f_0(1)}\right| > 1\right]. \quad (5.12)$$

Therefore, we have $\sigma_P^2/\sigma_N^2 < f_0(1)^2$. Given a bandlimited white Gaussian signal of power σ_P^2 , we can use (5.11) to calculate an approximate PWM clipping probability (upper bound). Note that if we use $\mathcal{ISL}^{L,1}$ in the calculation, we call this method 1. Similarly, if we use $\mathcal{ISL}^{L,2}$ in the calculation, we call this method 2. In Fig. 9, we plot the curves to show the relationship between the term $\sigma_P^2/(\sigma_N^2 \cdot f_0(1)^2)$ (shown in horizontal axis) and the clipping probability. Both method 1 and method 2 are applied with each considering three window sizes, i.e., $T = 2$, $T = 3$ and $T = 4$. The term σ_P^2/σ_N^2 is significantly larger than 0.637^2 for most reasonable clipping probabilities. Note that 0.637 is the coefficient that we derived in the sampling theorem. With method 2 and $T = 4$, the term σ_P^2/σ_N^2 is around 86% of $f_0(1)^2$, which is 0.6549 , for a clipping probability of 10^{-6} .

B. Lower Than PWM Frequency Sinusoids

The previous parts of this paper all considered the PWM frequency ($1/T$) to be the same as the Nyquist frequency of the input bandlimited signal. This subsection considers the PWM representation of those sinusoid (single tone) signals whose fundamental frequencies are lower than the predetermined PWM frequency. The analysis of PWM representation of sinusoids, particularly the spectral structure of the corresponding PWM waveform, is very popular in the literature. The single tone property of the sinusoids can lead to nice expressions for the spectrum of the natural sampling PWM signal or uniform sampling PWM signal [6], [27], [28]. However, low-pass recovery of either natural sampling PWM signals or uniform sampling PWM signals results in distortion of the modulating signal. This subsection shows that these lower than PWM frequency sinusoid signals can be represented as PWM waveforms exactly even when they have higher peaks than the peak constraint of our sampling theorem. Mathematically, consider the PWM representation of $x(t) = A_o \sin(2\pi f_o t + \pi/2)$. The samples of this sinusoid are taken at the predetermined PWM frequency which is higher than the fundamental frequency of this sinusoid. Suppose that these samples include at least some, if not all, peaks of the sinusoid. Let us first consider three-level PWM representation of two specific sinusoids for example.

1) *DC Signal* ($f_o = 0$): Consider the dc signal, e.g., $s(t) = 1$, which is an extreme case of the lower than PWM frequency sinusoids with fundamental frequency to be zero. It is apparent that the corresponding PWM waveform is also a dc signal, e.g., $pwm(t) = 1$. Thus, for the t th sample, we have $s_t = f_0(1) + \mathcal{ISI}_t = 1$. Clearly, the ISI to any sample is a constant, i.e., $\mathcal{ISI}_t = 1 - f_0(1) = 0.127$.

2) *Half PWM Frequency Sinusoid* ($f_o = 1/2T$): Consider the sinusoid whose frequency is half of the predetermined PWM frequency. The corresponding PWM waveform has widths $[\cdots 1 \ 0 \ -1 \ 0 \ 1 \ 0 \ -1 \ \cdots]$. The ISI to the pulses with width 1 has the same magnitude as the ISI to the pulses with width -1 ,

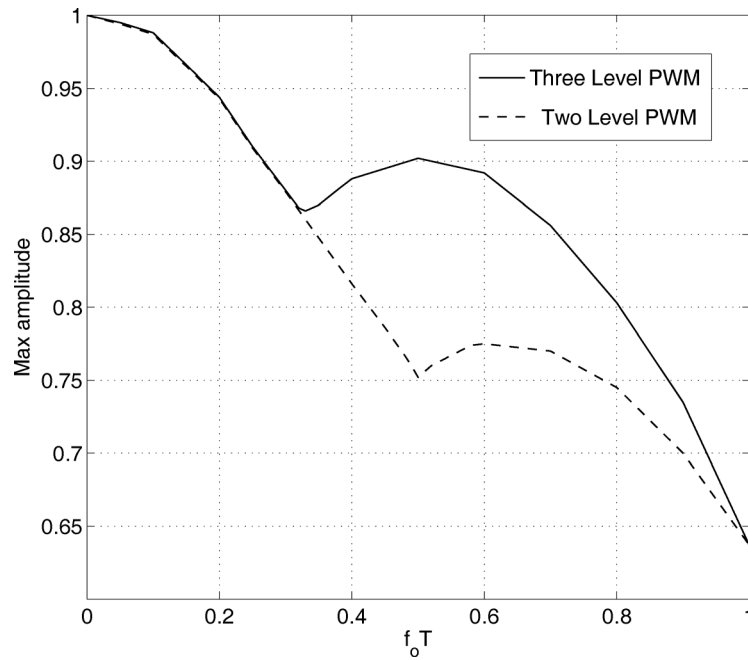


Fig. 10. Amplitude constraints for lower than PWM frequency sinusoids.

e.g., $|\mathcal{IS}\mathcal{I}_t| = 2 \sum_{n=1}^{+\infty} (-1)^{n+1} \cdot |f_{2n}(1)|$. The magnitude of the half PWM frequency sinusoid is $f_0(1) + |\mathcal{IS}\mathcal{I}_t| = 0.9003$. The ISI is actually helping during low-pass recovery since it has the same sign as the pulse.

These two examples of sinusoids have peaks higher than the peak constraint of the sampling theorem, but they can be exactly represented by PWM waveforms. For a fixed frequency of the input sinusoid, clearly there is a constraint on A_o of the sinusoid that exact PWM representation can support. However, it is very difficult to find the analytical solution for the allowable amplitude as a function of frequency. Fig. 10 shows the results of simulation for two-level and three-level PWM. It plots the maximum amplitudes of the input sinusoid, which allow for perfect PWM representation, versus the ratio ($f_o T$) of the sinusoid frequency to the PWM frequency. When the ratio ($f_o T$) is less than 0.33 (approximate value), the results for two-level and three-level PWM representations overlap with each other. When the ratio is between 0.33 and 1, three-level PWM representation can support higher amplitudes than two-level PWM representation. For a fixed fractional frequency, a sinusoid with smaller amplitude than the corresponding constraint illustrated in Fig. 10 can always be exactly represented by a PWM waveform with the corresponding level, while a sinusoid of larger amplitude cannot be represented exactly.

VI. CONCLUSION

Nyquist sampling is the classic way of representing a bandlimited signal. This paper has shown that a bandlimited waveform can be equally well represented by replacing each (variable amplitude) Nyquist sample with a (variable width) constant amplitude pulse. The proof is constructive and shows how to derive the pulsewidths from the Nyquist sample amplitudes. The only condition that must be satisfied is a peak constraint on the Nyquist samples that is a function of the amplitude of

the pulsewidth modulated waveform. Using a two-level waveform to represent a continuous waveform has great advantages for power efficient amplification because a saturating amplifier will not distort it. Furthermore, there is no need to create precise analog amplitudes; all that is needed is an accurate clock to set the pulsewidths. Applications range from audio to radio frequencies and power levels from milliwatts to hundreds of megawatts.

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